# Camera protected-power wiring / S1

Additive connection sheet. All source and destination ports below are functional until exact hardware identification and mapping.

**S1 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 10 Nov 2026**

**W202** **TB_POS.BUS -> F3.IN**

PACK_PROTECTED | WIRE | Protected pack; exact voltage envelope TBD

Separate F3/BUCK3/OV3 branch; select ratings before wiring

**W203** **F3.OUT -> BUCK3.IN+**

CAMERA_INPUT | WIRE | Protected pack; exact voltage envelope TBD

Separate F3/BUCK3/OV3 branch; select ratings before wiring

**W204** **TB_GND.BUS -> BUCK3.IN-**

GND | WIRE | 0 V reference

Separate F3/BUCK3/OV3 branch; select ratings before wiring

**W205** **BUCK3.IN- -> BUCK3.OUT-**

GND | INTERNAL_VERIFY | 0 V reference

Inspect existing bond only; no added conductor

**W206** **BUCK3.OUT+ -> OV3.IN+**

V5_CAMERA_RAW | WIRE | 5.00V target; unqualified

Separate F3/BUCK3/OV3 branch; select ratings before wiring

**W207** **BUCK3.OUT- -> OV3.GND**

GND | WIRE | 0 V reference

Separate F3/BUCK3/OV3 branch; select ratings before wiring

**W208** **OV3.OUT+ -> CAM1.5V**

V5_CAMERA | WIRE | Qualified 5V envelope; actual board limits TBD

Separate F3/BUCK3/OV3 branch; select ratings before wiring

**W209** **BUCK3.OUT- -> CAM1.GND**

GND | WIRE | 0 V reference

Separate F3/BUCK3/OV3 branch; select ratings before wiring

**W210** **OV3.OUT+ -> C_CAM.1**

V5_CAMERA | WIRE | Qualified 5V envelope; actual board limits TBD

Separate F3/BUCK3/OV3 branch; select ratings before wiring

**W211** **CAM1.GND -> C_CAM.2**

GND | WIRE | 0 V reference

Separate F3/BUCK3/OV3 branch; select ratings before wiring

BUCK3 IN-/OUT- bond W205 is an internal continuity inspection, never an extra jumper. Camera power return is dedicated to this branch. No physical ESP1 GPIO connection to CAM1. Laptop camera connection is Wi-Fi only; programmer is a separate temporary isolated setup.

Provisional allocation only: 5V x 1A = 5W continuous; 5V x 2A = 10W transient envelope. At assumed 80% efficiency: 6.25W / 12.5W input; losses 1.25W / 2.5W. These are design assumptions, not camera measurements or regulator qualification. Fuse current/thermal values remain TBD.

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.